NEERAJ KUMAR SINGH

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ABOUT

Applied AI / LLM engineer with 4+ years across NLP, explainable AI, model interpretability, and production-oriented RAG systems. Publication-backed researcher who ships Python/PyTorch/FastAPI tooling for LLM attribution, retrieval evaluation, agent workflows, and trustworthy AI.

EXPERIENCE

Lexsi AI, Mumbai, India
Research Scientist *April 2024 – March 2026*

- Shipped DLBacktrace v2, a production-grade explainability stack for relevance tracing and visualization across NLP and LLM workloads.
- Expanded transformer/LLM coverage via PyTorch `torch.export`/FX graphs; fixed attention relevance propagation and added LLaMA-family validation utilities.
- Added MoE relevance for JetMoE, OLMoE, Qwen3-MoE, and GPT-OSS; compressed attribution traces 17–93× with MAE < 0.001.
- Built `dlbacktrace-viz`, ModelRegistry, HF-compatible sampling/beam search, caching, and compressed trace storage for scalable attribution workflows.

CVPR Unit, Indian Statistical Institute, Kolkata
Research Assistant, NLP Lab *September 2021 – February 2024*

- Top contributor to "TrustED: Evaluating Trustworthiness of Deep Learning Systems"; developed explainable Indic hate-speech detection and intensity-reduction systems.
- Built a web application integrating tCNNs, Precily, and Paccmann models for IC50 prediction using drug and tissue-specific cell-line inputs.

SELECTED PROJECTS

- **XAI-RAG** – Production RAG with ChromaDB, Elasticsearch, RRF, BGE re-ranking, retrieval attribution, DeBERTa NLI faithfulness checks, RAGAS, OpenTelemetry, Redis, and async FastAPI/SSE endpoints.
- **Indic Research Agent** – Chainlit research assistant using LangGraph, LiteLLM, `query-kit`, PostgreSQL, Redis caching, streamed progress updates, and typed public-research tools.
- **query-kit** – Provider-agnostic Python CLI/library for ACL Anthology, arXiv, PubMed, Semantic Scholar, and OpenReview with normalized text/JSON results and sync/async APIs.
- **Explainable Agentic RAG** – LangGraph workflow with typed tools, structured outputs, retrieval attribution, verifier/retry loops, human review, tracing, and RAG quality metrics.

SELECTED PUBLICATIONS

- **DLBacktrace**: model-agnostic interpretability for MLPs, LLMs, and custom DNNs; benchmarked against SHAP, LIME, and Integrated Gradients. [Accepted]
- **XAI-Evals**: Python framework for local explanation evaluation with faithfulness, sensitivity, and robustness metrics. [arXiv]
- **SafeSpeech**: Indic hate-speech mitigation pipeline validated on five languages with BERTScores **0.96–0.99**. [Accepted]

TECHNICAL SKILLS

- **LLMs / NLP:** Transformers, BERT, T5, LLaMA, Qwen, Mixture-of-Experts, prompt/tool workflows, Hugging Face
- **RAG / Agents:** RAG, LangGraph, LangChain, hybrid search, embeddings, re-ranking, retrieval attribution, RAGAS
- **Backend / Data:** Python, FastAPI, async APIs, Chainlit, PostgreSQL, Redis, ChromaDB, Elasticsearch, Docker Compose, Git
- **XAI / ML:** DLBacktrace, `xai_evals`, SHAP, LIME, Integrated Gradients, PyTorch, TensorFlow, DeepSpeed, scikit-learn

EDUCATION

Indian Statistical Institute, Kolkata – M.Tech *2019 – 2021*
Bundelkhand Institute of Engineering & Technology, Jhansi – B.Tech *2015 – 2019*